Modelling Climate and Energy Futures: the course in three goes

Course structure, version 2 (23 October 2026). This replaces the single 15-week sequence with three passes ("goes") over the same subject. Each go covers the whole system, at a deeper level than the one before.

Go	Name	Question it answers	Learner at the end	Length (suggested)
1	**See the whole picture**	What is an integrated assessment model, and how do I read what it says?	Explains what goes into an IAM, what comes out, and how scenarios are built and used; tests each idea in an interactive explorer	6 modules, about 3 weeks
2	**Open the boxes**	How does each component work, and how do the components push on each other?	Describes each GCAM component, its main rule and its links to the others; builds a toy version of each in R or Python; runs and queries GCAM	8 modules, about 6 weeks
3	**Research with GCAM: transport**	How do I design, build and defend a research study with GCAM?	Reads and changes GCAM's transport sector, designs scenarios with uncertainty analysis, and delivers a reproducible study	8 modules, about 6 weeks plus project

The three goes can be taught as one 15-week course (3 + 6 + 6 weeks), as three separate short courses, or as a self-paced path. Each go has its own entry point, so a learner can start at Go 2 or Go 3 by passing that go's entry check.

Why three goes (re-evaluation of the 15-week version)

The 15-week version introduced every idea once, in depth, in a fixed order. Three problems showed up when the materials were reviewed:

1. **No early big picture.** Students met GCAM only in week 7, so weeks 1 to 6 felt like separate primers with no destination.
2. **Components taught in isolation.** Energy, land, water and policy each had a week, but the feedbacks between them (the reason IAMs exist) were mostly implicit.
3. **Research skills compressed into the end.** Uncertainty, model comparison, gcamdata and the project all landed in weeks 10 to 15, together with the hardest R.

Three goes fix this by spiral learning: every component appears three times, first as a box in a picture, then as a working mechanism with its connections, then as something the learner changes for research.

Topic	Go 1: see it	Go 2: open it	Go 3: research it
Climate	Emissions to warming in one chain; carbon budget	Hector inside GCAM; forcing; non-CO2 gases	Climate targets as constraints; emulators; uncertainty
Energy	Energy chain and Kaya	Supply, electricity, demand sectors, technology choice	Full transport sector, its data and calibration
Economics	Prices, carbon price, discounting	Markets, solver, logit choice, calibration	Elasticities, value of time, cost assumptions, sensitivity
Land and water	Why bioenergy needs land	Land nesting, yields, land carbon, water basins	Land and water side effects of transport fuels
Scenarios	SSPs, RCPs, AR6 categories	SSP quantification in GCAM; reference vs policy	Scenario design, ensembles, scenario discovery
Policy	What a carbon price and a net-zero target do	How GCAM implements prices, constraints, standards	Transport policy portfolios; net-zero transport
Tools	Browser explorers only	R or Python toy models; run GCAM; rgcam/gcamreader	gcamdata, add-on XML, batch runs, downscaling, coupling

Go 1: See the whole picture

Who it is for: anyone; no coding, no maths beyond percentages. **Style:** one big visual per idea, an everyday analogy, then an interactive explorer where the learner predicts first and then checks. **Materials:** Go 1 study guide (visual, with "predict then check" activities), browser explorers, short concept quizzes, the Go 1 mind map.

Module	Title	Big idea	Interactive check	From the 15-week version
1.1	What an IAM is	A model is a consistent "what if" machine, not a crystal ball	IAM machine: click each box to see its inputs and outputs	Week 1
1.2	The climate chain	Warming follows the running total of CO2	Warming explorer; carbon budget bathtub	Week 2
1.3	Energy and emissions	CO2 = people × income × energy per $ × CO2 per energy	Kaya calculator	Week 3
1.4	Prices and choices	Prices balance markets; a carbon price changes who wins; discounting weighs the future	Toy market; discounting explorer	Weeks 4 and 5 (concepts only)
1.5	Scenarios	SSPs tell stories about society, RCPs set the climate outcome, AR6 sorts pathways by warming; the new CMIP7 scenarios (2026) are what comes next	SSP × RCP matrix; scenario chart reader	Week 6
1.6	One model, end to end	A tour of GCAM: regions, markets, land, water, climate; reference vs	Policy lever explorer (price vs limit)	Weeks 7, 10, 12 and 14 (concepts only)

Module	Title	Big idea	Interactive check	From the 15-week version
		policy run; what IAMs cannot tell us		

Exit check: 20-question concept quiz (pass 70 %) and one “read a scenario figure” exercise.

Go 2: Open the boxes

Who it is for: learners who finished Go 1 (or pass its exit check). **Style:** each module opens one component: what it represents, the rule it follows, the numbers it needs, the components it talks to, a toy version built in 20 to 40 lines of R or Python, then where to see it in a real GCAM run. **Materials:** Go 2 component handbook, Go 2 implementation kit (R and Python toy models, GCAM query scripts), interaction maps, quizzes.

Module	Component	Toy implementation	GCAM implementation	From the 15-week version
2.1	Socioeconomic drivers	Grow population and GDP for an SSP; Kaya decomposition	Read GDP and population; compare SSPs	Weeks 3 and 6, labs 3 and 6
2.2	Markets, equilibrium and solving	Solve one market, then three linked markets	How GCAM’s marketplace and solver work; reading solver logs	Week 4 and 7, lab 4
2.3	Choice: the logit	Share three technologies by cost; nest two levels	Share weights and exponents in input files	Week 7, lab 7
2.4	Energy supply	Levelised cost; resource supply curve; learning curve	Electricity mix and a cheap-solar add-on	Week 8, lab 8
2.5	Energy demand: buildings, industry, transport	Service demand with income elasticity; mode choice with time cost	Final energy and service output by sector	Week 8
2.6	Land, food and bioenergy	Land sharing by profit; land for bioenergy	Land allocation and land-use change emissions	Week 9, lab 9
2.7	Water	Basin supply	Water	Week 9, lab 9

Module	Component	Toy implementation	GCAM implementation	From the 15-week version
		curve; water for power cooling	withdrawals by basin and sector	
2.8	Emissions, climate and policy: the system together	Hector two paths; carbon price vs cap; LP contrast	Reference vs carbon price vs net zero; cross-component effects	Weeks 2, 10, 11 and 12, labs 2, 10, 11, 12

Each module ends with an **interaction map**: which components send it information (prices, quantities, emissions) and which it sends to. Module 2.8 joins all eight maps into one and traces five feedback loops through a real GCAM run.

Exit check: component quiz (one question set per module), a toy-model notebook that runs end to end, and a two-page note explaining one cross-component feedback with GCAM output.

Go 3: Research with GCAM, a transport case

Who it is for: learners who finished Go 2 and want to publish or advise with GCAM. **Style:** research handbook. Detailed, referenced, with implementation code, worked research designs and a literature review. The transport sector is the running case. **Materials:** Go 3 research handbook, transport research kit (add-on XML templates, query file, R and Python analysis scripts, scenario-design templates), reading pack, project brief and rubric.

Module	Title	Content	From the 15-week version
3.1	GCAM as research software	Architecture, configuration, input XML, calibration, solver diagnostics, version control, reproducibility	Weeks 7, 10 and 12, labs 7, 10 and 12
3.2	Transport in GCAM, in full	Passenger and freight structure, nests, value of time, load factors, costs, intensities, elasticities, calibration, data sources	New
3.3	What the literature knows	IAM transport studies with GCAM and other models; model comparison; gaps	New, builds on week 12

Module	Title	Content	From the 15-week version
3.4	Designing a study	Research questions, scenario logic, SSPs, policy design, uncertainty plan	Weeks 10 and 13
3.5	Building transport scenarios	Add-on XML for vehicle costs, load factors, share weights, fuels; carbon price, constraints, standards; batch runs	Labs 8, 10 and 13
3.6	Analysing and stress-testing results	Queries, decomposition, sensitivity, ensembles, scenario discovery, comparison with AR6	Labs 12 and 13
3.7	Going finer and wider	Regional focus (Saudi Arabia, Middle East), downscaling (DSCALE), coupling with LEAP, MESSAGEix-Transport and stock models	New
3.8	Limits, ethics and writing it up	Behavioural realism, demand-side options, equity, carbon removal, community guidelines for GCAM papers, IAMC reporting	Weeks 14 and 15

Exit: a research project with a written report, reproducible code and a presentation (rubric in the Go 3 handbook).

Assessment across the three goes

Component	Go 1	Go 2	Go 3
Concept quizzes	20 %	10 %	
Explorer and “predict then	10 %		

Component	Go 1	Go 2	Go 3
check” sheets			
Toy-model notebooks		15 %	
Feedback note with GCAM output		10 %	
Research proposal			5 %
Project report and code			20 %
Project presentation			10 %

Weights are a suggestion for a single 15-week course (total 100 %). Taught separately, each go’s components scale to 100 %.

What carries over from the existing materials

- **Lab scripts 1 to 14** and the GCAM files stay; they are re-used in Go 2 (labs 2 to 11) and Go 3 (labs 7, 10, 12 and 13).
- **The 15 weekly decks** remain valid as lecture material; Go 1 uses the first half of weeks 1 to 7, Go 2 uses weeks 2 to 12, and Go 3 adds new transport material.
- **The study guides and mind maps** for the three stages remain as background reading; the new guides for each go are the primary reading.
- **The interactive explorers** on the course site feed Go 1 and Go 2 directly.

Software versions (checked 23 October 2026)

- GCAM: latest release 9.1 (2 June 2026). The online documentation describes v8.2; the course follows the documentation for structure and notes where 9.x differs.
- Reading GCAM output: gcamreader 1.5.0 (Python, `pip install gcamreader`), rgcam, rchart and rmap (R, from GitHub).
- Climate: hector R package 3.2.0 (GitHub) and FaIR 2.2.5 (Python, `pip install fair`).
- Toy models for Go 2 run in base R (4.x) or Python 3 with numpy, scipy and matplotlib.
- gcamextractor and gcamland were archived in May 2026, and pygcam only supports GCAM 5 and 6, so the course does not use them.